

Title

House Price Prediction using Linear Regression

Objective

To build a machine learning model that predicts house prices.

Tools Used

- Python
- Pandas
- Scikit-Learn
- Matplotlib

Methodology

1. Loaded the dataset.
2. Selected important numerical features.
3. Split data into training and testing sets.
4. Trained a Linear Regression model.
5. Evaluated the model using MAE, MSE and R^2 Score.

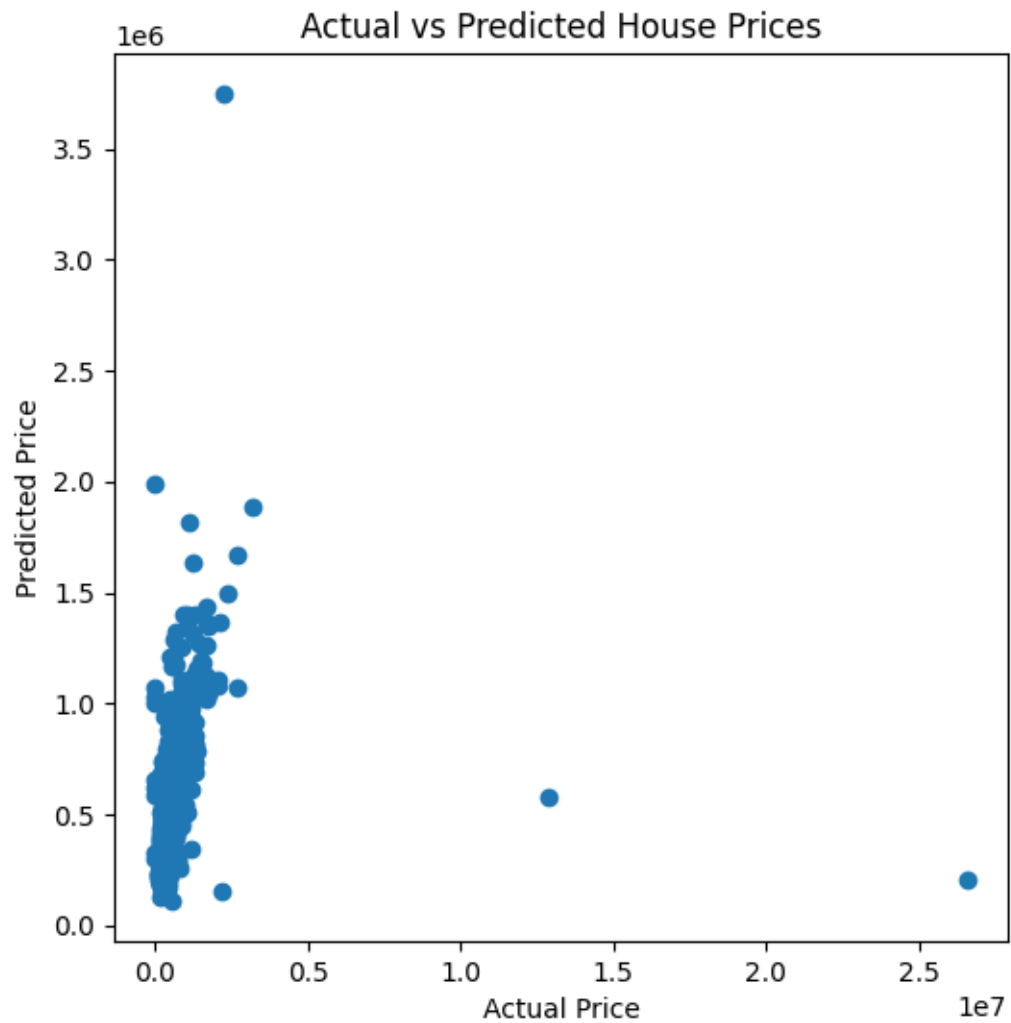
Results

MAE : 210907.95

MSE : 98691835089.62

R^2 Score : 0.0323

Insert:



```
PS C:\Users\hr815\OneDrive\Desktop\Internship\Task-2-House-Price-Prediction> python house_price_prediction.py
Dataset Shape: (4600, 18)
   date           price  bedrooms  bathrooms  ...      street      city  statezip  country
0  2014-05-02 00:00:00   313000.0        3.0        1.50  ...  18810 Densmore Ave N  Shoreline  WA  98133    USA
1  2014-05-02 00:00:00  2384000.0        5.0        2.50  ...    709 W Blaine St   Seattle  WA  98119    USA
2  2014-05-02 00:00:00   342000.0        3.0        2.00  ...  26206-26214 143rd Ave SE    Kent  WA  98042    USA
3  2014-05-02 00:00:00   420000.0        3.0        2.25  ...    857 170th Pl NE  Bellevue  WA  98008    USA
4  2014-05-02 00:00:00   550000.0        4.0        2.50  ...   9105 170th Ave NE   Redmond  WA  98052    USA

[5 rows x 18 columns]

Model Performance
MAE : 210907.9548851089
MSE : 986918350899.619
R2 Score : 0.03228720631044091
PS C:\Users\hr815\OneDrive\Desktop\Internship\Task-2-House-Price-Prediction>
```

Conclusion

A Linear Regression model was successfully developed and tested for house price prediction.